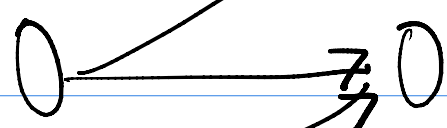


Network - ex 1

1.A.1

University =

Students Courses



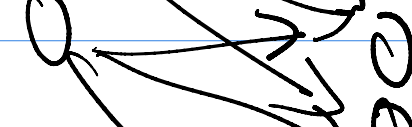
↳ could be how much credit / grade

wikipedia hyperlinks = Page References



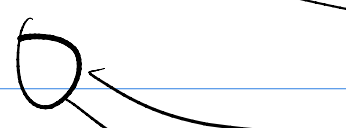
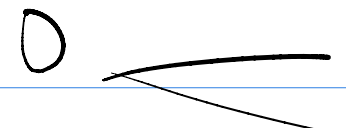
↳ could be the number of revision

Interbank lending = Bank Borrower



↳ How much money is being credited

Power transmission grid = Power grid Villages / town / homes



↳ Undirected since AC power goes both way

↳ Electricity usage (in kwh)

Co authorship in research = authors



↳ undirected

↳ number of joint papers

In university enrollment a plain list of students cannot tell the whole story regarding which students performs best at which subjects, whether a subject is "easy" judging by their average grades.

1.A.2

1. The information that is missing is the relationship between all of the students to determine who interacts with who.

2. 2 types of edges,

"is a friend with"

undirected

helps show potential paths

"communicated with today"

directed

more precise but narrower in possibility

3. Look for a node in which most outward paths originate from. Because an initiator of a fast spreading rumor is usually positioned to reach many disjoint part of the node.

1.B.1

1. App A has the most likelihood to survive since it already has a dense interconnected network. New user will find their friends already there
2. Because network A's user is not an individual nodes but they're a dense interconnected network. If some individual from network A (high-degree) recommends the app. It'll reach many others simultaneously.
3. Because if a product has the advantage of having a highly dense interconnected user base as described in (1) then it'll have the inherent advantage of growing to be more popular in the future just based on the users alone.

1.B.2 Vaccinating a slow disease: Question: whos the most important to vaccinate first?

How: By adopting centrality

Detecting informal team in log: Question = Are there groups

How = Community detection (Louvain)

Critical router failure -- Question: How robust?

How = Connectivity detection

Modeling viral tweet: Question = How does it spread?

How: Using diffusion models

Community idea exchange: Question: Groups

How: Bridges

1.C.1

1. Scenario A will reach more students first than since it will spread through nodes with more degrees.

2. Yes, in a connected network the rumor will eventually spread to all of

the students. the condition is called connectivity. with any starting point it will reach all students.

3. Who receives the message first is deterministic for the speed of the spread. If it reach a high edge node it'll reach to more people faster.